

Must-Answer DBMS Interview Questions

DBMS Fundamentals

1. What is a database?
2. What is a DBMS?
3. What is the difference between data and information?
4. Why do we need a DBMS?
5. What are the main advantages of using a DBMS?
6. What are the disadvantages of using a DBMS?
7. What is the difference between DBMS and RDBMS?
8. What is a database system?
9. What are the components of a database system?
10. What is metadata?
11. What is a data dictionary?
12. What is a database schema?
13. What is a database instance?
14. What is the difference between schema and instance?
15. What is data independence?
16. What is physical data independence?
17. What is logical data independence?
18. Why is data independence important?
19. What is data abstraction?
20. What are the different levels of data abstraction?
21. What is the view level in DBMS?
22. What is the logical level in DBMS?
23. What is the physical level in DBMS?
24. What is data redundancy?
25. What is data inconsistency?
26. How does DBMS reduce data redundancy?
27. What is data integrity?
28. What is data security in DBMS?
29. What is data isolation?
30. What is concurrent access?
31. What is a database application?
32. What is a database administrator?
33. What are the responsibilities of a DBA?
34. What is a database designer?
35. What is an end user in DBMS?
36. What are the types of database users?
37. What is a file processing system?
38. What are the limitations of file processing systems?
39. How is DBMS better than a file processing system?
40. What is a centralized database?
41. What is a distributed database?

42. What is a federated database?
43. What is a cloud database?
44. What is an in-memory database?
45. What is an object-oriented database?
46. What is a hierarchical database model?
47. What is a network database model?
48. What is a relational database model?
49. What is a NoSQL database?
50. What are the major types of NoSQL databases?

Data Models

51. What is a data model?
52. Why are data models used?
53. What are the different types of data models?
54. What is a conceptual data model?
55. What is a logical data model?
56. What is a physical data model?
57. What is the entity-relationship model?
58. What is the relational model?
59. What is the object-based data model?
60. What is a semi-structured data model?
61. What is the difference between conceptual and logical design?
62. What is the difference between logical and physical design?
63. What is a record-based data model?
64. What is a graph data model?
65. What is a document data model?

ER Model

66. What is an entity?
67. What is an entity set?
68. What is an attribute?
69. What is a relationship?
70. What is a relationship set?
71. What is an ER diagram?
72. Why are ER diagrams used?
73. What is a strong entity?
74. What is a weak entity?
75. What is an identifying relationship?
76. What is a partial key?
77. What is a simple attribute?
78. What is a composite attribute?
79. What is a single-valued attribute?
80. What is a multi-valued attribute?
81. What is a derived attribute?
82. What is a stored attribute?

83. What is a key attribute?
84. What is a null attribute?
85. What is cardinality in an ER diagram?
86. What is participation constraint?
87. What is total participation?
88. What is partial participation?
89. What is a one-to-one relationship?
90. What is a one-to-many relationship?
91. What is a many-to-one relationship?
92. What is a many-to-many relationship?
93. What is a recursive relationship?
94. What is a ternary relationship?
95. What is the degree of a relationship?
96. What is specialization?
97. What is generalization?
98. What is aggregation?
99. What is inheritance in ER modeling?
100. What is an ISA relationship?
101. What is disjoint specialization?
102. What is overlapping specialization?
103. What is total specialization?
104. What is partial specialization?
105. How do you convert an ER diagram into relational tables?
106. How is a weak entity converted into a relational table?
107. How is a multi-valued attribute converted into a relational table?
108. How is a one-to-one relationship mapped into tables?
109. How is a one-to-many relationship mapped into tables?
110. How is a many-to-many relationship mapped into tables?
111. What mistakes should be avoided while designing an ER diagram?
112. How would you design an ER diagram for a library management system?
113. How would you design an ER diagram for a hospital management system?
114. How would you design an ER diagram for a student course registration system?
115. How would you design an ER diagram for an e-commerce system?

Relational Model

116. What is a relation?
117. What is a tuple?
118. What is an attribute?
119. What is a domain?
120. What is a relational schema?
121. What is a relational instance?
122. What is the degree of a relation?
123. What is the cardinality of a relation?
124. What is a table in the relational model?
125. What is a row in the relational model?
126. What is a column in the relational model?

127. What is an atomic value?
128. What are the properties of a relation?
129. Why are duplicate rows not allowed in a relation?
130. Why is the order of rows insignificant in a relation?
131. Why is the order of columns insignificant in a relation?
132. What is a super key?
133. What is a candidate key?
134. What is a primary key?
135. What is an alternate key?
136. What is a foreign key?
137. What is a composite key?
138. What is a compound key?
139. What is a surrogate key?
140. What is a natural key?
141. What is a unique key?
142. What is the difference between primary key and unique key?
143. What is the difference between primary key and foreign key?
144. Can a primary key contain null values?
145. Can a foreign key contain null values?
146. Can a table have multiple candidate keys?
147. Can a table have multiple primary keys?
148. Can a table have multiple foreign keys?
149. What is referential integrity?
150. What is entity integrity?
151. What is domain integrity?
152. What is user-defined integrity?
153. What are integrity constraints?
154. What is a check constraint?
155. What is a default constraint?
156. What is a not null constraint?
157. What is an assertion in DBMS?
158. What is a view in the relational model?
159. What is a base table?
160. What is relational algebra?
161. What is relational calculus?

Functional Dependencies

162. What is a functional dependency?
163. What is a trivial functional dependency?
164. What is a non-trivial functional dependency?
165. What is a completely non-trivial functional dependency?
166. What is a partial dependency?
167. What is a full functional dependency?
168. What is a transitive dependency?
169. What is a multivalued dependency?
170. What is a join dependency?

- 171. What is closure of a set of attributes?
- 172. What is closure of a set of functional dependencies?
- 173. What are Armstrong's axioms?
- 174. What is reflexivity rule?
- 175. What is augmentation rule?
- 176. What is transitivity rule?
- 177. What is union rule?
- 178. What is decomposition rule?
- 179. What is pseudotransitivity rule?
- 180. How do you find candidate keys using functional dependencies?
- 181. What is an extraneous attribute?
- 182. What is a canonical cover?
- 183. What is a minimal cover?
- 184. How do you find a minimal cover?
- 185. Why are functional dependencies important in normalization?

Normalization

- 186. What is normalization?
- 187. Why is normalization needed?
- 188. What are insertion anomalies?
- 189. What are deletion anomalies?
- 190. What are update anomalies?
- 191. What is denormalization?
- 192. When should denormalization be used?
- 193. What is 1NF?
- 194. What is 2NF?
- 195. What is 3NF?
- 196. What is BCNF?
- 197. What is 4NF?
- 198. What is 5NF?
- 199. What is 6NF?
- 200. What is the difference between 1NF and 2NF?
- 201. What is the difference between 2NF and 3NF?
- 202. What is the difference between 3NF and BCNF?
- 203. Why is BCNF stricter than 3NF?
- 204. Can a relation be in 3NF but not in BCNF?
- 205. Can a relation be in BCNF but not in 3NF?
- 206. What is dependency preservation?
- 207. What is lossless decomposition?
- 208. What is lossy decomposition?
- 209. How do you check whether a decomposition is lossless?
- 210. How do you check whether a decomposition is dependency preserving?
- 211. What is decomposition in DBMS?
- 212. Why is decomposition used?
- 213. What is the chase test?
- 214. What are the disadvantages of excessive normalization?

- 215. How do you normalize a table step by step?
- 216. How do you identify the highest normal form of a relation?
- 217. What is prime attribute?
- 218. What is non-prime attribute?
- 219. How are candidate keys related to normalization?
- 220. How are functional dependencies related to normalization?

SQL Basics

- 221. What is SQL?
- 222. What are the different categories of SQL commands?
- 223. What is DDL?
- 224. What is DML?
- 225. What is DCL?
- 226. What is TCL?
- 227. What is DQL?
- 228. What is the difference between SQL and MySQL?
- 229. What is the difference between SQL and PL/SQL?
- 230. What is the difference between **CHAR** and **VARCHAR**?
- 231. What is the difference between **VARCHAR** and **TEXT**?
- 232. What is the difference between **DATE**, **TIME**, **DATETIME**, and **TIMESTAMP**?
- 233. What is the difference between **INT**, **FLOAT**, **DOUBLE**, and **DECIMAL**?
- 234. What is the purpose of **CREATE**?
- 235. What is the purpose of **ALTER**?
- 236. What is the purpose of **DROP**?
- 237. What is the purpose of **TRUNCATE**?
- 238. What is the purpose of **DELETE**?
- 239. What is the difference between **DELETE**, **DROP**, and **TRUNCATE**?
- 240. What is the difference between **WHERE** and **HAVING**?
- 241. What is the difference between **GROUP BY** and **ORDER BY**?
- 242. What is the purpose of **DISTINCT**?
- 243. What is the purpose of **LIMIT**?
- 244. What is the purpose of **OFFSET**?
- 245. What is the purpose of **BETWEEN**?
- 246. What is the purpose of **IN**?
- 247. What is the purpose of **LIKE**?
- 248. What is the purpose of **IS NULL**?
- 249. What is the difference between **NULL**, zero, and empty string?
- 250. What is three-valued logic in SQL?
- 251. How does SQL handle comparisons with **NULL**?
- 252. What is the purpose of aliases in SQL?
- 253. What is the difference between **COUNT(*)** and **COUNT(column)**?
- 254. What is the difference between **UNION** and **UNION ALL**?
- 255. What is the difference between **INTERSECT** and **EXCEPT**?
- 256. What is a subquery?
- 257. What is a correlated subquery?
- 258. What is a nested query?

- 259. What is a scalar subquery?
- 260. What is the difference between **ANY** and **ALL**?
- 261. What is the difference between **EXISTS** and **IN**?
- 262. What is the difference between **NOT EXISTS** and **NOT IN**?
- 263. What is a common table expression?
- 264. What is a recursive CTE?
- 265. What is a window function?
- 266. What is the difference between aggregate functions and window functions?
- 267. What is the purpose of **ROW_NUMBER()**?
- 268. What is the purpose of **RANK()**?
- 269. What is the purpose of **DENSE_RANK()**?
- 270. What is the difference between **RANK()** and **DENSE_RANK()**?
- 271. What is the purpose of **PARTITION BY**?
- 272. What is the purpose of **OVER()**?
- 273. What is the purpose of **CASE** in SQL?
- 274. What is the purpose of **COALESCE()**?
- 275. What is the purpose of **NULLIF()**?

SQL Joins

- 276. What is a join?
- 277. Why are joins used?
- 278. What is an inner join?
- 279. What is a left join?
- 280. What is a right join?
- 281. What is a full outer join?
- 282. What is a cross join?
- 283. What is a self join?
- 284. What is a natural join?
- 285. What is an equi join?
- 286. What is a non-equi join?
- 287. What is an anti join?
- 288. What is a semi join?
- 289. What is the difference between inner join and outer join?
- 290. What is the difference between left join and right join?
- 291. What is the difference between cross join and inner join?
- 292. What is the difference between self join and recursive query?
- 293. What is the difference between join condition in **ON** and filter condition in **WHERE**?
- 294. How can a left join behave like an inner join?
- 295. How do joins handle duplicate rows?
- 296. How do joins handle null values?
- 297. How do you find unmatched rows using joins?
- 298. How do you find duplicate records using SQL?
- 299. How do you find the second highest salary using SQL?
- 300. How do you find the nth highest salary using SQL?
- 301. How do you fetch employees with salary greater than their department average?
- 302. How do you fetch departments having more than a given number of employees?

- 303. How do you fetch records present in one table but not in another?
- 304. How do you delete duplicate rows from a table?
- 305. How do you update one table using values from another table?

Advanced SQL Querying

- 306. How do you write a query to find duplicate values in a column?
- 307. How do you write a query to find unique values in a column?
- 308. How do you find records with missing values?
- 309. How do you find customers who placed no orders?
- 310. How do you find employees who do not belong to any department?
- 311. How do you find the top 3 salaries in each department?
- 312. How do you calculate running total in SQL?
- 313. How do you calculate moving average in SQL?
- 314. How do you find gaps in sequential data?
- 315. How do you find consecutive login days?
- 316. How do you pivot rows into columns?
- 317. How do you unpivot columns into rows?
- 318. How do you get monthly sales from daily sales data?
- 319. How do you calculate percentage contribution using SQL?
- 320. How do you find the first order of each customer?
- 321. How do you find the latest record for each group?
- 322. How do you find employees earning more than their managers?
- 323. How do you fetch hierarchical data using SQL?
- 324. How do you implement pagination in SQL?
- 325. How do you optimize a slow SQL query?
- 326. How do you read an execution plan?
- 327. What is query cost?
- 328. What is selectivity?
- 329. What is cardinality estimation?
- 330. What are statistics in SQL databases?

Views

- 331. What is a view?
- 332. Why are views used?
- 333. What is a simple view?
- 334. What is a complex view?
- 335. What is a materialized view?
- 336. What is the difference between a view and a materialized view?
- 337. Can we update a view?
- 338. What are the conditions for an updatable view?
- 339. What are the advantages of views?
- 340. What are the disadvantages of views?
- 341. How can views improve security?
- 342. Can views improve performance?
- 343. What happens when a base table used by a view is dropped?

344. What is **WITH CHECK OPTION**?

345. What is an indexed view?

Indexing

346. What is an index?

347. Why are indexes used?

348. How does an index improve query performance?

349. What are the disadvantages of indexes?

350. What is a primary index?

351. What is a secondary index?

352. What is a clustered index?

353. What is a non-clustered index?

354. What is the difference between clustered and non-clustered indexes?

355. Can a table have multiple clustered indexes?

356. Can a table have multiple non-clustered indexes?

357. What is a dense index?

358. What is a sparse index?

359. What is the difference between dense index and sparse index?

360. What is a composite index?

361. What is a covering index?

362. What is a unique index?

363. What is a hash index?

364. What is a bitmap index?

365. What is a B-tree index?

366. What is a B+ tree index?

367. Why are B+ trees commonly used in databases?

368. What is the difference between B-tree and B+ tree?

369. What is an index scan?

370. What is an index seek?

371. What is a full table scan?

372. When should indexes be created?

373. When should indexes be avoided?

374. How do indexes affect insert, update, and delete operations?

375. What is the leftmost prefix rule in composite indexes?

376. What is index fragmentation?

377. What is index reorganization?

378. What is index rebuilding?

379. How do you choose columns for indexing?

380. Why may an index not be used by the query optimizer?

Transactions

381. What is a transaction?

382. Why are transactions important?

383. What are ACID properties?

384. What is atomicity?

- 385. What is consistency?
- 386. What is isolation?
- 387. What is durability?
- 388. What is a transaction state?
- 389. What are the states of a transaction?
- 390. What is an active transaction?
- 391. What is a partially committed transaction?
- 392. What is a committed transaction?
- 393. What is a failed transaction?
- 394. What is an aborted transaction?
- 395. What is a terminated transaction?
- 396. What is **COMMIT**?
- 397. What is **ROLLBACK**?
- 398. What is **SAVEPOINT**?
- 399. What is autocommit?
- 400. What is a dirty read?
- 401. What is a non-repeatable read?
- 402. What is a phantom read?
- 403. What is a lost update?
- 404. What is a write skew problem?
- 405. What are isolation levels?
- 406. What is read uncommitted isolation?
- 407. What is read committed isolation?
- 408. What is repeatable read isolation?
- 409. What is serializable isolation?
- 410. What is snapshot isolation?
- 411. What is the difference between repeatable read and serializable?
- 412. What is the difference between read committed and repeatable read?
- 413. What is transaction schedule?
- 414. What is a serial schedule?
- 415. What is a non-serial schedule?
- 416. What is a serializable schedule?
- 417. What is conflict serializability?
- 418. What is view serializability?
- 419. What is recoverable schedule?
- 420. What is cascadeless schedule?
- 421. What is strict schedule?
- 422. What is cascading rollback?
- 423. What is two-phase commit?
- 424. What is distributed transaction?
- 425. How do databases ensure durability?

Concurrency Control

- 426. What is concurrency control?
- 427. Why is concurrency control needed?
- 428. What are the problems caused by concurrent transactions?

- 429. What is locking?
- 430. What is a shared lock?
- 431. What is an exclusive lock?
- 432. What is an update lock?
- 433. What is an intention lock?
- 434. What is lock compatibility?
- 435. What is lock granularity?
- 436. What is row-level locking?
- 437. What is table-level locking?
- 438. What is page-level locking?
- 439. What is two-phase locking?
- 440. What is strict two-phase locking?
- 441. What is rigorous two-phase locking?
- 442. What is conservative two-phase locking?
- 443. What is timestamp ordering protocol?
- 444. What is optimistic concurrency control?
- 445. What is pessimistic concurrency control?
- 446. What is multiversion concurrency control?
- 447. What is MVCC?
- 448. How does MVCC work?
- 449. What is a read lock?
- 450. What is a write lock?
- 451. What is lock escalation?
- 452. What is starvation in DBMS?
- 453. What is deadlock?
- 454. What are the necessary conditions for deadlock?
- 455. How can deadlocks be prevented?
- 456. How can deadlocks be detected?
- 457. How can deadlocks be recovered from?
- 458. What is wait-die scheme?
- 459. What is wound-wait scheme?
- 460. What is a wait-for graph?

Recovery System

- 461. What is database recovery?
- 462. Why is recovery needed in DBMS?
- 463. What are the types of failures in DBMS?
- 464. What is transaction failure?
- 465. What is system crash?
- 466. What is disk failure?
- 467. What is media failure?
- 468. What is log-based recovery?
- 469. What is a transaction log?
- 470. What is write-ahead logging?
- 471. What is redo operation?
- 472. What is undo operation?

- 473. What is checkpointing?
- 474. Why are checkpoints used?
- 475. What is shadow paging?
- 476. What is immediate database modification?
- 477. What is deferred database modification?
- 478. What is ARIES recovery algorithm?
- 479. What is a dirty page?
- 480. What is a buffer pool?
- 481. What is a steal policy?
- 482. What is a no-steal policy?
- 483. What is a force policy?
- 484. What is a no-force policy?
- 485. What is backup?
- 486. What is restore?
- 487. What is full backup?
- 488. What is incremental backup?
- 489. What is differential backup?
- 490. What is point-in-time recovery?

Storage and File Organization

- 491. What is physical storage in DBMS?
- 492. What is a disk block?
- 493. What is a page in DBMS?
- 494. What is a record?
- 495. What is fixed-length record storage?
- 496. What is variable-length record storage?
- 497. What is file organization?
- 498. What is heap file organization?
- 499. What is sequential file organization?
- 500. What is hash file organization?
- 501. What is clustered file organization?
- 502. What is B+ tree file organization?
- 503. What is hashing in DBMS?
- 504. What is static hashing?
- 505. What is dynamic hashing?
- 506. What is extendible hashing?
- 507. What is linear hashing?
- 508. What is a hash collision?
- 509. What is collision resolution?
- 510. What is a bucket?
- 511. What is an overflow bucket?
- 512. What is buffer management?
- 513. What is buffer replacement?
- 514. What is LRU replacement policy?
- 515. What is MRU replacement policy?
- 516. What is clock replacement policy?

- 517. What is sequential access?
- 518. What is random access?
- 519. What is RAID?
- 520. What are the different RAID levels?

Query Processing and Optimization

- 521. What is query processing?
- 522. What are the steps in query processing?
- 523. What is query parsing?
- 524. What is query translation?
- 525. What is query optimization?
- 526. What is query execution?
- 527. What is a query execution plan?
- 528. What is a logical query plan?
- 529. What is a physical query plan?
- 530. What is a query optimizer?
- 531. What is rule-based optimization?
- 532. What is cost-based optimization?
- 533. What is heuristic optimization?
- 534. What is query rewriting?
- 535. What is predicate pushdown?
- 536. What is projection pushdown?
- 537. What is join ordering?
- 538. Why does join order matter?
- 539. What is nested loop join?
- 540. What is block nested loop join?
- 541. What is indexed nested loop join?
- 542. What is merge join?
- 543. What is hash join?
- 544. What is the difference between hash join and merge join?
- 545. What is the difference between nested loop join and hash join?
- 546. When is a full table scan better than an index scan?
- 547. What is pipelining in query execution?
- 548. What is materialization in query execution?
- 549. What are database statistics?
- 550. How can stale statistics affect query performance?

Relational Algebra and Calculus

- 551. What is relational algebra?
- 552. What are the basic operations of relational algebra?
- 553. What is selection operation?
- 554. What is projection operation?
- 555. What is union operation?
- 556. What is set difference operation?
- 557. What is Cartesian product?

- 558. What is rename operation?
- 559. What is join operation?
- 560. What is division operation?
- 561. What is intersection operation?
- 562. What is natural join in relational algebra?
- 563. What is theta join?
- 564. What is equi join?
- 565. What is outer join in relational algebra?
- 566. What is tuple relational calculus?
- 567. What is domain relational calculus?
- 568. What is the difference between relational algebra and relational calculus?
- 569. What is procedural query language?
- 570. What is non-procedural query language?
- 571. What is a safe expression in relational calculus?
- 572. What is a range variable?
- 573. How do you express SQL queries using relational algebra?
- 574. How do you express joins using relational algebra?
- 575. How do you express division using relational algebra?

Database Design

- 576. What is database design?
- 577. What are the phases of database design?
- 578. What is requirement analysis in database design?
- 579. What is conceptual database design?
- 580. What is logical database design?
- 581. What is physical database design?
- 582. What is schema design?
- 583. What is schema refinement?
- 584. What is top-down database design?
- 585. What is bottom-up database design?
- 586. What is data modeling?
- 587. What is domain analysis?
- 588. What is a business rule in database design?
- 589. How do you choose primary keys during design?
- 590. How do you model optional relationships?
- 591. How do you model many-to-many relationships?
- 592. How do you model hierarchical data?
- 593. How do you model historical data?
- 594. How do you model audit logs?
- 595. How do you design a database for scalability?
- 596. How do you design a database for reporting?
- 597. How do you design a database for high write volume?
- 598. How do you design a database for high read volume?
- 599. How do you avoid data duplication during design?
- 600. How do you decide between normalization and denormalization?

Distributed Databases

- 601. What is a distributed database?
- 602. What is a distributed DBMS?
- 603. What are the advantages of distributed databases?
- 604. What are the disadvantages of distributed databases?
- 605. What is data fragmentation?
- 606. What is horizontal fragmentation?
- 607. What is vertical fragmentation?
- 608. What is mixed fragmentation?
- 609. What is data replication?
- 610. What is full replication?
- 611. What is partial replication?
- 612. What is data allocation?
- 613. What is location transparency?
- 614. What is fragmentation transparency?
- 615. What is replication transparency?
- 616. What is distributed query processing?
- 617. What is distributed transaction management?
- 618. What is two-phase commit in distributed databases?
- 619. What is three-phase commit?
- 620. What is distributed deadlock?
- 621. What is distributed concurrency control?
- 622. What is eventual consistency?
- 623. What is strong consistency?
- 624. What is causal consistency?
- 625. What is read-your-writes consistency?
- 626. What is CAP theorem?
- 627. What does consistency mean in CAP theorem?
- 628. What does availability mean in CAP theorem?
- 629. What does partition tolerance mean in CAP theorem?
- 630. Can a distributed system provide all three CAP properties at once?

NoSQL and Modern Databases

- 631. What is NoSQL?
- 632. Why were NoSQL databases introduced?
- 633. What are the advantages of NoSQL databases?
- 634. What are the disadvantages of NoSQL databases?
- 635. What is a key-value database?
- 636. What is a document database?
- 637. What is a column-family database?
- 638. What is a graph database?
- 639. What is the difference between SQL and NoSQL databases?
- 640. What is schema-less design?
- 641. What is flexible schema?
- 642. What is BASE property?

- 643. What is eventual consistency in NoSQL?
- 644. What is sharding?
- 645. What is horizontal scaling?
- 646. What is vertical scaling?
- 647. What is replication in NoSQL?
- 648. What is partitioning in NoSQL?
- 649. What is a replica set?
- 650. What is a collection in MongoDB?
- 651. What is a document in MongoDB?
- 652. What is an embedded document?
- 653. What is referencing in document databases?
- 654. When should you embed data in MongoDB?
- 655. When should you reference data in MongoDB?
- 656. What is Cassandra?
- 657. What is Redis?
- 658. What is Neo4j?
- 659. What is Elasticsearch?
- 660. When should you choose NoSQL over SQL?

Security and Authorization

- 661. What is database security?
- 662. Why is database security important?
- 663. What is authentication?
- 664. What is authorization?
- 665. What is access control?
- 666. What is role-based access control?
- 667. What is discretionary access control?
- 668. What is mandatory access control?
- 669. What is a database role?
- 670. What is a privilege?
- 671. What is **GRANT**?
- 672. What is **REVOKE**?
- 673. What is SQL injection?
- 674. How can SQL injection be prevented?
- 675. What are prepared statements?
- 676. What are parameterized queries?
- 677. What is input validation?
- 678. What is database encryption?
- 679. What is encryption at rest?
- 680. What is encryption in transit?
- 681. What is data masking?
- 682. What is auditing in DBMS?
- 683. What is database backup security?
- 684. What is least privilege principle?
- 685. What is privilege escalation?

Triggers, Procedures, and Functions

- 686. What is a stored procedure?
- 687. What is a function in SQL?
- 688. What is the difference between a stored procedure and a function?
- 689. What is a trigger?
- 690. Why are triggers used?
- 691. What are the types of triggers?
- 692. What is a before trigger?
- 693. What is an after trigger?
- 694. What is an instead-of trigger?
- 695. What is a row-level trigger?
- 696. What is a statement-level trigger?
- 697. What are the advantages of triggers?
- 698. What are the disadvantages of triggers?
- 699. Can a trigger call a stored procedure?
- 700. What is a cursor?
- 701. What are the types of cursors?
- 702. What is an implicit cursor?
- 703. What is an explicit cursor?
- 704. What are the disadvantages of cursors?
- 705. When should cursors be avoided?

Practical Placement Questions

- 706. Explain DBMS to a non-technical person.
- 707. Explain normalization with a real-life example.
- 708. Explain indexing with a real-life example.
- 709. Explain transactions using an ATM example.
- 710. Explain ACID properties using a bank transfer example.
- 711. Explain primary key and foreign key using a college database example.
- 712. Explain joins using a student-course example.
- 713. Explain deadlock using a real-life example.
- 714. Explain the difference between OLTP and OLAP.
- 715. Explain when you would use SQL and when you would use NoSQL.
- 716. Design a database for a placement management system.
- 717. Design a database for an online food delivery system.
- 718. Design a database for an online shopping platform.
- 719. Design a database for a movie ticket booking system.
- 720. Design a database for a banking system.
- 721. Design a database for a railway reservation system.
- 722. Design a database for a hotel booking system.
- 723. Design a database for a social media application.
- 724. Design a database for a ride-sharing application.
- 725. Design a database for an inventory management system.
- 726. How would you store passwords securely in a database?
- 727. How would you handle millions of records in a table?

- 728. How would you improve a slow search query?
- 729. How would you design tables for user login and roles?
- 730. How would you prevent duplicate email registration?
- 731. How would you track order status history?
- 732. How would you store many phone numbers for one user?
- 733. How would you store tags for blog posts?
- 734. How would you model followers in a social media app?
- 735. How would you model likes and comments?
- 736. How would you model product categories and subcategories?
- 737. How would you model cart and orders in e-commerce?
- 738. How would you maintain stock quantity during concurrent orders?
- 739. How would you handle cancellation and refund records?
- 740. How would you handle soft deletes?
- 741. How would you handle audit columns like created_at and updated_at?
- 742. How would you handle schema changes in production?
- 743. How would you migrate data from one schema to another?
- 744. How would you take backup and restore a production database?
- 745. How would you decide which columns to index?

Short Difference-Based Questions

- 746. What is the difference between DBMS and RDBMS?
- 747. What is the difference between schema and instance?
- 748. What is the difference between primary key and candidate key?
- 749. What is the difference between candidate key and super key?
- 750. What is the difference between candidate key and alternate key?
- 751. What is the difference between natural key and surrogate key?
- 752. What is the difference between foreign key and reference key?
- 753. What is the difference between entity and attribute?
- 754. What is the difference between entity and entity set?
- 755. What is the difference between relationship and relationship set?
- 756. What is the difference between weak entity and strong entity?
- 757. What is the difference between total participation and partial participation?
- 758. What is the difference between specialization and generalization?
- 759. What is the difference between aggregation and composition?
- 760. What is the difference between 3NF and BCNF?
- 761. What is the difference between normalization and denormalization?
- 762. What is the difference between lossless and lossy decomposition?
- 763. What is the difference between dependency preservation and lossless decomposition?
- 764. What is the difference between clustered and non-clustered index?
- 765. What is the difference between dense and sparse index?
- 766. What is the difference between B-tree and B+ tree?
- 767. What is the difference between hash indexing and tree indexing?
- 768. What is the difference between primary index and secondary index?
- 769. What is the difference between table scan and index scan?
- 770. What is the difference between index scan and index seek?
- 771. What is the difference between **DELETE** and **TRUNCATE**?

- 772. What is the difference between **DROP** and **TRUNCATE**?
- 773. What is the difference between **WHERE** and **HAVING**?
- 774. What is the difference between **GROUP BY** and **ORDER BY**?
- 775. What is the difference between **UNION** and **UNION ALL**?
- 776. What is the difference between **IN** and **EXISTS**?
- 777. What is the difference between **JOIN** and subquery?
- 778. What is the difference between correlated and non-correlated subquery?
- 779. What is the difference between stored procedure and function?
- 780. What is the difference between trigger and stored procedure?
- 781. What is the difference between view and table?
- 782. What is the difference between view and materialized view?
- 783. What is the difference between OLTP and OLAP?
- 784. What is the difference between data warehouse and database?
- 785. What is the difference between data lake and data warehouse?
- 786. What is the difference between SQL and NoSQL?
- 787. What is the difference between vertical scaling and horizontal scaling?
- 788. What is the difference between sharding and partitioning?
- 789. What is the difference between replication and backup?
- 790. What is the difference between strong consistency and eventual consistency?

Rapid-Fire One-Line Questions

- 791. What does ACID stand for?
- 792. What does SQL stand for?
- 793. What does DDL stand for?
- 794. What does DML stand for?
- 795. What does DCL stand for?
- 796. What does TCL stand for?
- 797. What does ER stand for?
- 798. What does DBA stand for?
- 799. What does OLTP stand for?
- 800. What does OLAP stand for?
- 801. What does CAP stand for?
- 802. What does BASE stand for?
- 803. Which normal form removes partial dependency?
- 804. Which normal form removes transitive dependency?
- 805. Which normal form handles multivalued dependency?
- 806. Which SQL command is used to remove all rows from a table quickly?
- 807. Which SQL command is used to remove a table structure?
- 808. Which SQL command is used to undo a transaction?
- 809. Which SQL command is used to make a transaction permanent?
- 810. Which constraint prevents null values?
- 811. Which constraint ensures uniqueness?
- 812. Which constraint links two tables?
- 813. Which join returns matching rows from both tables?
- 814. Which join returns all rows from the left table?
- 815. Which join returns all combinations of rows?

- 816. Which index stores data rows in sorted order?
- 817. Which tree structure is commonly used for database indexing?
- 818. Which isolation level prevents dirty reads?
- 819. Which isolation level provides the strongest isolation?
- 820. Which concurrency problem occurs when uncommitted data is read?
- 821. Which concurrency problem occurs when the same row gives different values in the same transaction?
- 822. Which concurrency problem occurs when new rows appear during repeated reads?
- 823. Which protocol has growing and shrinking phases?
- 824. Which graph is used for deadlock detection?
- 825. Which recovery rule says logs must be written before data pages?

Scenario-Based Questions

- 826. A table has repeating groups in one column; which normal form is violated?
- 827. A table has a non-prime attribute depending on part of a composite key; which normal form is violated?
- 828. A table has a non-prime attribute depending on another non-prime attribute; which normal form is violated?
- 829. A relation is in 3NF but not BCNF; what kind of dependency usually causes this?
- 830. A query is slow even after indexing; what factors would you check?
- 831. An index improves SELECT queries but slows INSERT queries; why?
- 832. A transaction reads uncommitted data from another transaction; what problem is this?
- 833. Two transactions wait for each other's locks; what problem is this?
- 834. A database crashes after commit but before data is written to disk; which ACID property matters?
- 835. A user tries to insert a child row without a parent row; which constraint fails?
- 836. A column should accept only values greater than zero; which constraint should be used?
- 837. A relation needs to store multiple skills for one employee; how should it be modeled?
- 838. A many-to-many relationship exists between students and courses; how should it be implemented?
- 839. An application needs fast lookup by email; what database feature should be considered?
- 840. A report query scans millions of rows every time; what optimizations could be considered?
- 841. A table contains duplicate customer records; how would you detect them?
- 842. A database must support high availability across regions; what design choices matter?
- 843. A system needs strict bank-transfer correctness; which database properties matter most?
- 844. A social media feed needs massive scale and flexible schema; what database choices could be considered?
- 845. A product search page needs full-text search; what database feature or system could be used?

Interview Deep-Dive Questions

- 846. Why is the relational model still widely used?
- 847. Why are keys important in database design?
- 848. Why should every table usually have a primary key?
- 849. Why are foreign keys important?
- 850. Why can too many indexes hurt performance?
- 851. Why is normalization not always enough for real systems?
- 852. Why do databases use logs?
- 853. Why is serializability important?

- 854. Why is conflict serializability easier to check than view serializability?
- 855. Why is 2PL considered a concurrency control protocol?
- 856. Why can deadlocks still occur under 2PL?
- 857. Why is MVCC popular in modern databases?
- 858. Why can **NULL** cause confusion in SQL queries?
- 859. Why is **NOT IN** dangerous when null values are present?
- 860. Why should passwords not be stored in plain text?
- 861. Why are prepared statements safer than string concatenation?
- 862. Why are database backups not enough without restore testing?
- 863. Why does schema design affect application performance?
- 864. Why is query optimization important?
- 865. Why do distributed databases face consistency challenges?
- 866. Why does CAP theorem matter in system design?
- 867. Why is eventual consistency acceptable in some applications?
- 868. Why is strong consistency required in some applications?
- 869. Why is denormalization used in analytics systems?
- 870. Why are data warehouses different from transactional databases?

Company-Style SQL Practice Questions

- 871. Write a query to find the second highest salary.
- 872. Write a query to find the nth highest salary.
- 873. Write a query to find employees with the same salary.
- 874. Write a query to find duplicate emails.
- 875. Write a query to delete duplicate emails.
- 876. Write a query to find employees who joined in the last 30 days.
- 877. Write a query to count employees department-wise.
- 878. Write a query to find departments with no employees.
- 879. Write a query to find employees without managers.
- 880. Write a query to find managers with more than five employees.
- 881. Write a query to find the highest salary in each department.
- 882. Write a query to find the top three salaries in each department.
- 883. Write a query to find customers who never placed an order.
- 884. Write a query to find customers who placed more than three orders.
- 885. Write a query to find the latest order of each customer.
- 886. Write a query to find total revenue per month.
- 887. Write a query to find products never ordered.
- 888. Write a query to find best-selling products.
- 889. Write a query to find users who logged in consecutively for three days.
- 890. Write a query to find gaps in invoice numbers.
- 891. Write a query to swap two column values.
- 892. Write a query to update salary by 10% for a department.
- 893. Write a query to find common records between two tables.
- 894. Write a query to find records in table A but not in table B.
- 895. Write a query to rank students by marks.
- 896. Write a query to calculate cumulative sales.
- 897. Write a query to calculate percentage of total sales.

- 898. Write a query to fetch every alternate row.
- 899. Write a query to fetch only duplicate rows.
- 900. Write a query to fetch only unique rows.

Final Revision Checklist Questions

- 901. Can you explain DBMS architecture clearly?
- 902. Can you explain the three-schema architecture?
- 903. Can you explain ER diagram symbols?
- 904. Can you convert an ER diagram into tables?
- 905. Can you identify keys from a relation?
- 906. Can you find candidate keys using functional dependencies?
- 907. Can you compute attribute closure?
- 908. Can you find minimal cover?
- 909. Can you identify the highest normal form of a relation?
- 910. Can you decompose a relation into 3NF?
- 911. Can you decompose a relation into BCNF?
- 912. Can you check lossless decomposition?
- 913. Can you check dependency preservation?
- 914. Can you explain ACID with examples?
- 915. Can you identify transaction anomalies?
- 916. Can you explain isolation levels with examples?
- 917. Can you identify deadlock conditions?
- 918. Can you explain locking protocols?
- 919. Can you explain indexing with examples?
- 920. Can you choose the right index for a query?
- 921. Can you explain query optimization basics?
- 922. Can you read a simple query execution plan?
- 923. Can you write joins confidently?
- 924. Can you write subqueries confidently?
- 925. Can you write grouping queries confidently?
- 926. Can you write window function queries confidently?
- 927. Can you explain views and materialized views?
- 928. Can you explain triggers and stored procedures?
- 929. Can you explain backup and recovery?
- 930. Can you compare SQL and NoSQL clearly?
- 931. Can you explain CAP theorem simply?
- 932. Can you design a database for a real-world system?
- 933. Can you justify normalization choices?
- 934. Can you justify denormalization choices?
- 935. Can you explain database security basics?
- 936. Can you explain SQL injection prevention?
- 937. Can you handle common SQL interview problems?
- 938. Can you answer difference-based DBMS questions quickly?
- 939. Can you explain DBMS concepts with real-life examples?
- 940. Can you speak about your database project design decisions?